Green's Theorem: From Circulation to Winding and Residues

N-20260417

The local derivative of a vector field determines circulation around a contractible boundary. What happens when the field is defined on a region with a hole? The answer leads from Green's theorem to winding numbers and, after expressing a planar vector field as a complex differential, to the residue theorem. A single punctured-plane calculation will connect all three subjects.[1]

§1.1 Green's theorem is cancellation along internal edges

Let $D \subset \mathbb{R}^2$ be a bounded region with piecewise smooth boundary, oriented so that the region is on the left as one walks along each component. Suppose $P, Q \in C^1$ on a neighborhood of the closure of D. Green's theorem states

$$\boxed{\int_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dx\,dy.}$$

On a rectangle, the result follows by the one-variable fundamental theorem of calculus applied to opposite sides. Subdividing the region creates edges traversed once in each direction, so their contributions cancel. Outer boundary components are counterclockwise and boundary components enclosing holes are clockwise.

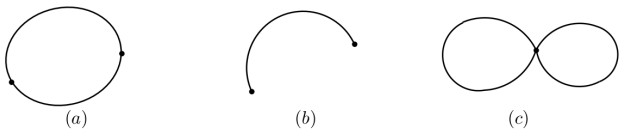


Figure 1.1: Boundary orientation distinguishes an enclosed region from an arbitrary traversal of a curve.

Writing $\omega = P\,dx + Q\,dy$, the differential is

$$d\omega = (Q_x - P_y)\,dx \wedge dy,$$

and Green's theorem becomes $\int_{\partial D} \omega = \int_D d\omega$. The identity $d^2 = 0$ is related to the cancellation principle: applying two consecutive boundary operations also gives zero. The diagrammatic cancellation has an algebraic counterpart.

§1.2 Curl zero is local; a potential is global

A field $F = (P, Q)$ has a potential ϕ when $\omega = d\phi$, or

$$P = \phi_x, \qquad Q = \phi_y.$$

[1] MIT 18.02SC (Green's theorem); MIT 18.04, *Complex Variables with Applications* (holomorphic forms).

Such a form is called exact. It satisfies $d\omega = d^2\phi = 0$, so $Q_x = P_y$. Conversely, on a simply connected open region U, a C^1 form with $d\omega = 0$ is exact.

One way to construct the potential is to choose a base point $x_0 \in U$ and define

$$\phi(x) = \int_{\gamma_{x_0,x}} \omega,$$

where $\gamma_{x_0,x}$ is a piecewise smooth path from x_0 to x. For a simply connected region, any two such paths are homotopic with fixed endpoints. Green's theorem, applied through a homotopy or by subdivision, makes their integrals agree when $d\omega = 0$. The resulting ϕ is well defined and has $d\phi = \omega$.

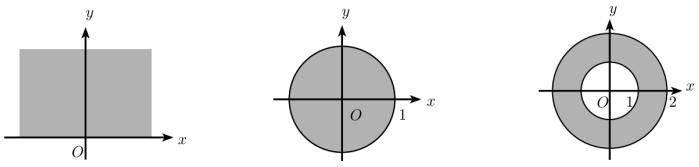


Figure 1.2: A loop around a hole need not bound a region contained in the domain of the field.

§1.3 The angular form detects winding

On $U = \mathbb{R}^2 \setminus \{0\}$, consider

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

A direct derivative calculation gives $d\alpha = 0$, but on the counterclockwise unit circle $\gamma(t) = (\cos t, \sin t)$,

$$\gamma^*\alpha = dt, \qquad \int_\gamma \alpha = 2\pi.$$

Thus no globally defined real potential can have differential α.

The value is topological. Any piecewise smooth closed curve γ avoiding the origin has a continuous polar angle along its parameter interval, say $\gamma(t) = r(t)(\cos\theta(t), \sin\theta(t))$ with $r(t) > 0$. Along the curve,

$$\gamma^*\alpha = d\theta, \qquad \int_\gamma \alpha = \theta(1) - \theta(0).$$

Because the curve is closed, the difference is an integer multiple of 2π. This integer is its winding number:

$$\boxed{\operatorname{wind}(\gamma, 0) = \frac{1}{2\pi}\int_\gamma \alpha \in \mathbb{Z}.}$$

It is invariant under deformations of the curve that avoid the origin.

§1.4 Cohomology of the Punctured Plane

Suppose ω is any smooth closed one-form on U. Let

$$c = \frac{1}{2\pi}\int_{S^1} \omega.$$

The form $\omega - c\alpha$ has zero integral around the unit circle. Every closed loop in U is homotopic, up to orientation and repeated traversal, to an integer number of such circles. Since integrals of closed forms are homotopy invariant, every closed-loop integral of $\omega - c\alpha$ is zero.

The same path-integral construction as above therefore gives a global smooth function f with

$$\boxed{\omega = df + c\alpha.}$$

The coefficient c is unique because α has nonzero period. Hence the first de Rham cohomology group is

$$H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{R},$$

with $[\alpha]$ as a generator.

This does more than say that the punctured plane has a hole. It computes exactly one independent obstruction to constructing a potential: every closed one-form differs from an exact one by a multiple of the angular form.

§1.5 Holomorphic one-forms lead to Cauchy and residues

Write $z = x + iy$. A complex-valued form $f(z)\,dz$, for $f = u + iv$, splits into real and imaginary parts:

$$f(z)\,dz = (u\,dx - v\,dy) + i(v\,dx + u\,dy).$$

If f is holomorphic, the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ show that both real one-forms are closed:

$$d(f\,dz) = 0.$$

Consequently, integrals of a holomorphic function around contractible loops vanish: this is Cauchy's theorem in the language of Green's theorem.

Near an isolated simple pole a, suppose

$$f(z) = \frac{c}{z-a} + h(z),$$

with h holomorphic. For a small positively oriented circle C around a,

$$\int_C f(z)\,dz = c\int_C \frac{dz}{z-a} = 2\pi i\,c.$$

Indeed $dz/(z-a)$ has imaginary part equal to the angular form centered at a, and its real part $d\log|z-a|$ has zero period. The coefficient c is the residue at a.

If a meromorphic function has finitely many poles in a bounded region and is holomorphic on the boundary, remove small disks around its poles. Cauchy's theorem on the punctured region, with clockwise inner boundary components, gives

$$\boxed{\int_{\partial D} f(z)\,dz = 2\pi i \sum_{a\in D} \mathrm{Res}(f,a).}$$

The residue theorem is therefore a local-to-global circulation law. Holomorphicity makes the exterior derivative zero away from singularities, while each isolated pole contributes a quantized boundary period.

The same geometry also computes area: choosing $P = -y/2$, $Q = x/2$ gives

$$\mathrm{area}(D) = \frac{1}{2}\int_{\partial D} (x\,dy - y\,dx).$$

In that formula the derivative is 1, so area is accumulated locally. In the winding and residue formulas the derivative vanishes away from singularities, so all the information comes from what a loop encloses. Green's theorem encompasses both phenomena through the orientation and topology of the boundary.